

James Smith

Mechatronics Engineering Student

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Education

Program Name, University Name

Expected Month Year

- Relevant coursework: Course A, Course B, Course C

Technical Skills

Languages

- C, C++, Python, MATLAB

Tools

- SolidWorks, KiCad, Git, ROS

Areas

- Embedded systems, control theory, PCB design, CAD

Experience

Mechatronics Intern, Example Robotics Co.

May 2025 – Aug 2025

- Designed and validated a vision-guided pick-and-place fixture, cutting cycle time by 12%.
- Reduced manual changeover steps from 6 to 2 through fixture redesign.

Projects

Autonomous Line-Following Robot

Jan 2026 – Mar 2026

- Designed a PID-controlled line-following robot with gain scheduling, improving lap times 38%.
- Fabricated and populated a custom PCB for the sensor array and motor drivers.

3-DOF Desktop Robotic Arm

Sep 2025 – Dec 2025

- Derived closed-form inverse kinematics for a 3-link planar arm.
- Iterated a compliant 3D-printed gripper jaw through four print revisions.

Leadership & Activities

Formula SAE — Electronics Lead

Sep 2025 – Present

- Lead a 4-person subteam designing the car's CAN bus telemetry network.